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Multimodal Chatbot for Plant Disease Recognition

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- Agriculture is a critical sector for global food security. Plant diseases can significantly impact crop yield and quality.
- Farmers often struggle to identify diseases promptly due to a lack of access to expert knowledge.
- A chatbot can bridge this gap by providing instant disease recognition and treatment suggestions.



Objectives

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The major objectives of the project are as follows:-

- Identify plant diseases from images of leaves.
- Provide text-based diagnosis and treatment suggestions.
- Retrieve relevant information from a knowledge base using Retrieval-Augmented Generation (RAG).
- Ensure fast and accurate responses with a FAISS-based vector search engine.



Literature Review (1/4)

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Reference	Year	Summarized Abstract	Dataset Used	Methods Used	Challenges
[1]	2020	DP-CycleGAN enhances quality of diseased leaf images using structure and texture loss. Improves recognition in data-scarce scenarios.	Maize leaf images (PlantVillage)	DP-CycleGAN, SP/TP loss, VGG16BN	Low data, annotation cost, overfitting, poor augmentation, no lesion guidance
[2]	2023	Discusses AI (ML/DL) applications for plant disease detection with IoT integration.	PlantVillage	ML, DL, IoT	Food demand, water scarcity, symptom detection, climate impact
[3]	2021	AI for tomato disease classification using C-GAN and DenseNet121.	PlantVillage	C-GAN, DenseNet121	Early detection, data quality, plant part variance



Literature Review (2/4)

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Reference	Year	Summarized Abstract	Dataset Used	Methods Used	Challenges
[4]	2018	Mobile ML recommender for real-time disease diagnosis in Uganda with 3939 QA pairs.	Cassava, maize, beans QA set	RetBERT, Seq2Seq, HayStack	Low internet access, power limits, multilingual handling
[5]	2024	CNNs trained on 87K images of 25 crops achieve 99.53% accuracy.	PlantVillage	CNN, AlexNet, VGG, Overfeat	Dataset overlap, complex conditions, shading, distractions
[6]	2024	AELGNet classifies medicinal leaves with 99.71% accuracy, beating 14 baselines.	Indian Medicinal Leaf dataset	AELGNet, MBConv, Adam	Global features, compute cost, lightweight demand



Literature Review (3/4)

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Reference	Year	Summarized Abstract	Dataset Used	Methods Used	Challenges
[7]	2024	Mobile crop app detects 21 leaf anomalies with 94% accuracy; supports 3 languages.	20,718 images (quinoa, cucumber, pepper, tomato)	CNN, ODK	Language barrier, high quality requirement, limited scope
[8]	2024	Achieves 94% accuracy on PlantVillage and 97% on new datasets.	87,848 leaf images	Inception, F1, precision	Environmental noise, imbalance, generalization difficulty
[9]	2024	Uses edges + color analysis + EfficientNetB3 for robust detection.	PlantVillage 2020	DenseNet201, EfficientNetB3	Lighting, leaf rotation, artifacts, occlusion



Literature Review (4/4)

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Reference	Year	Summarized Abstract	Dataset Used	Methods Used	Challenges
[10]	2023	MDSCIRNet classifies potato diseases with 99.24% using multi-head attention.	Potato Leaf Disease dataset	MDSCIRNet, ESRGAN, CLAHE	JPG loss, re-optimization, transparency in DL
[11]	2024	Real-time CNN-based tomato disease detection in Jalgaon region.	Jalgaon tomato fields	CNN, gradient descent, cross-entropy	Fresh data complexity, small samples, convergence issues
[12]	2024	CoffeeNet using ResNet-50 with attention gets 98.54% accuracy.	Arabica Coffee Leaf Dataset	ResNet-50, spatial-channel attention	Lighting, distortion, scale/size changes

Implementation using MobileNetV2

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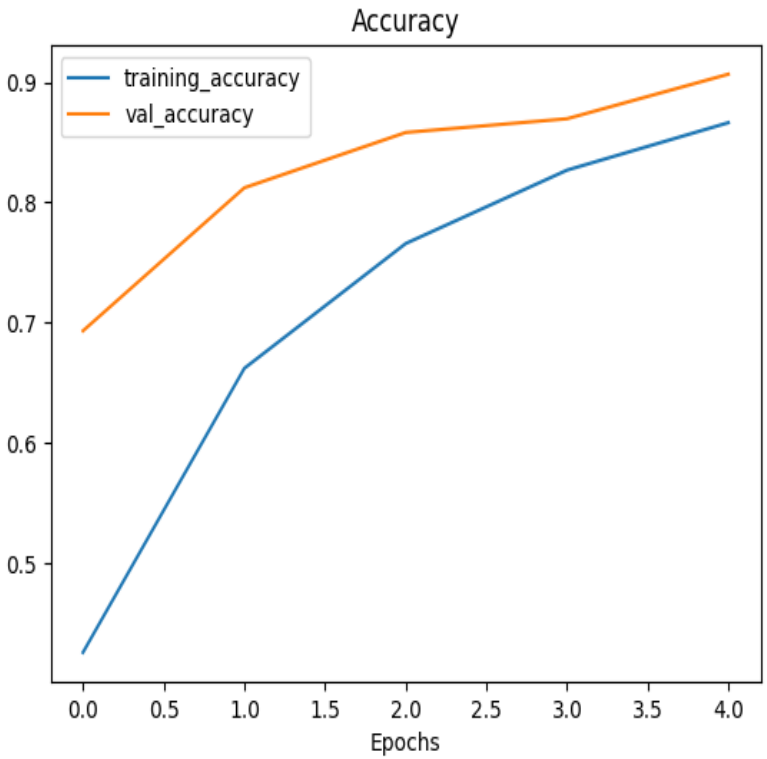
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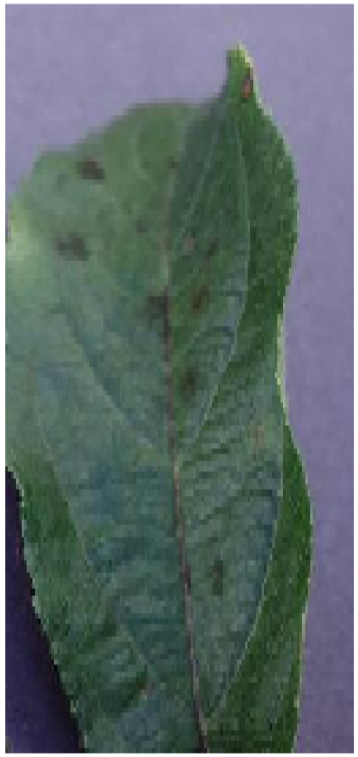
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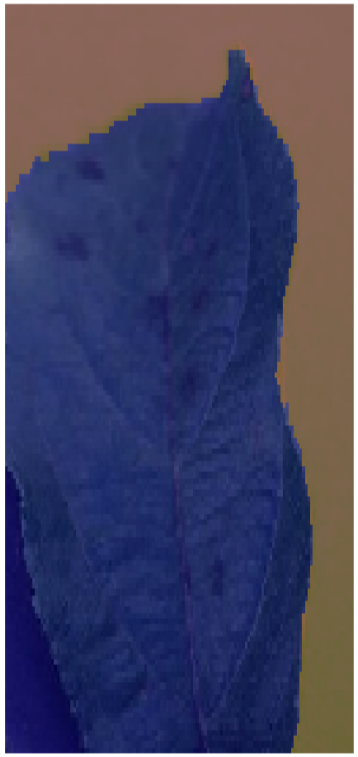




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Metric	Value	Remarks
Training Accuracy	98.2%	Indicates strong learning capability on the dataset.
Validation Accuracy	95.1%	Suggests good generalization on unseen data, with minor overfitting.
F1-Score	¿0.90 for most classes	High precision and recall across multiple disease classes.
Training Loss	0.12	Low loss demonstrates effective optimization during training.
Validation Loss	0.22	Slightly higher than training loss, showing generalization with some room for improvement.



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Components of the system:-

- Text-based disease retrieval: FAISS-powered semantic search for disease-related information.
- Image-based disease recognition: Deep learning model (CNN) classifies plant diseases.
- LLaMA 2 chatbot: Generates detailed responses based on retrieved knowledge.
- Django API: Handles user interaction and integration.



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Model: $\text{vit}_h(ViT - Huge)$

- Purpose: Automatically segment and highlight the infected region of the leaf image.
- Explanation: SAM can perform zero-shot segmentation of arbitrary objects. Given a point (e.g., center of the image), it generates a binary mask of the corresponding region. This helps localize the diseased area visually.

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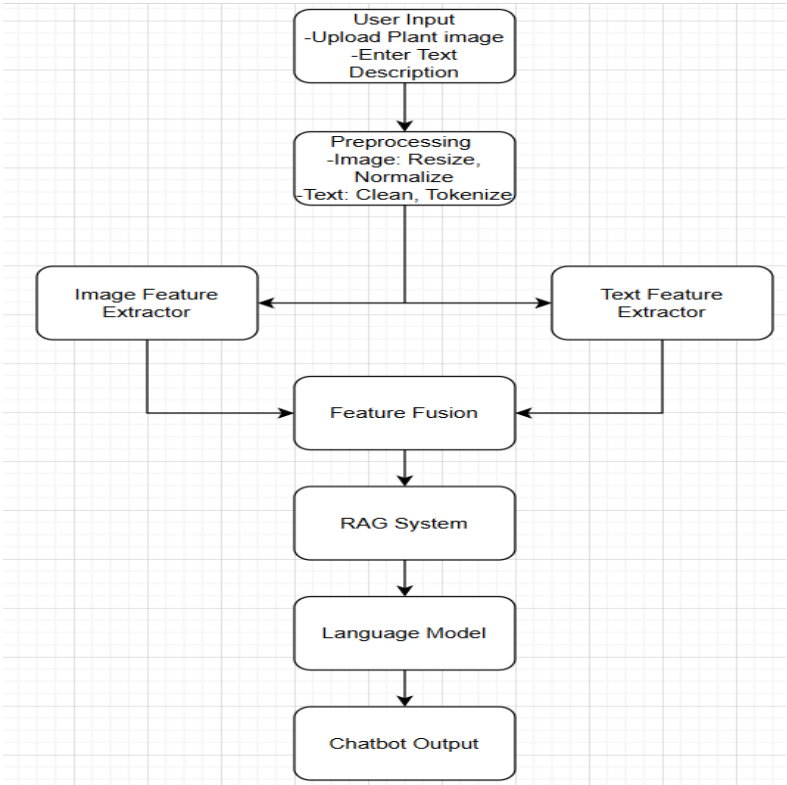
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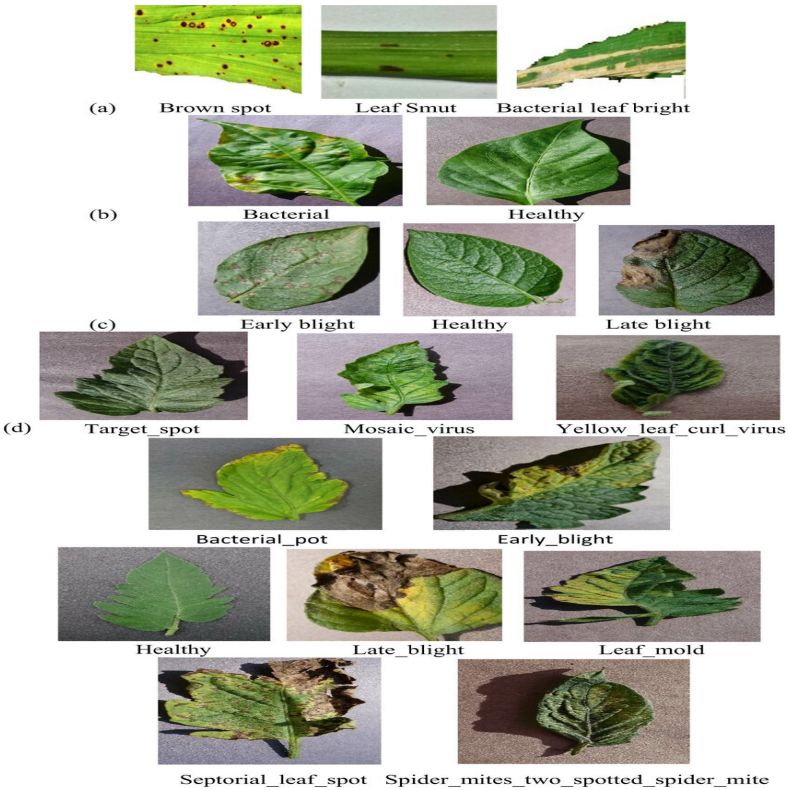
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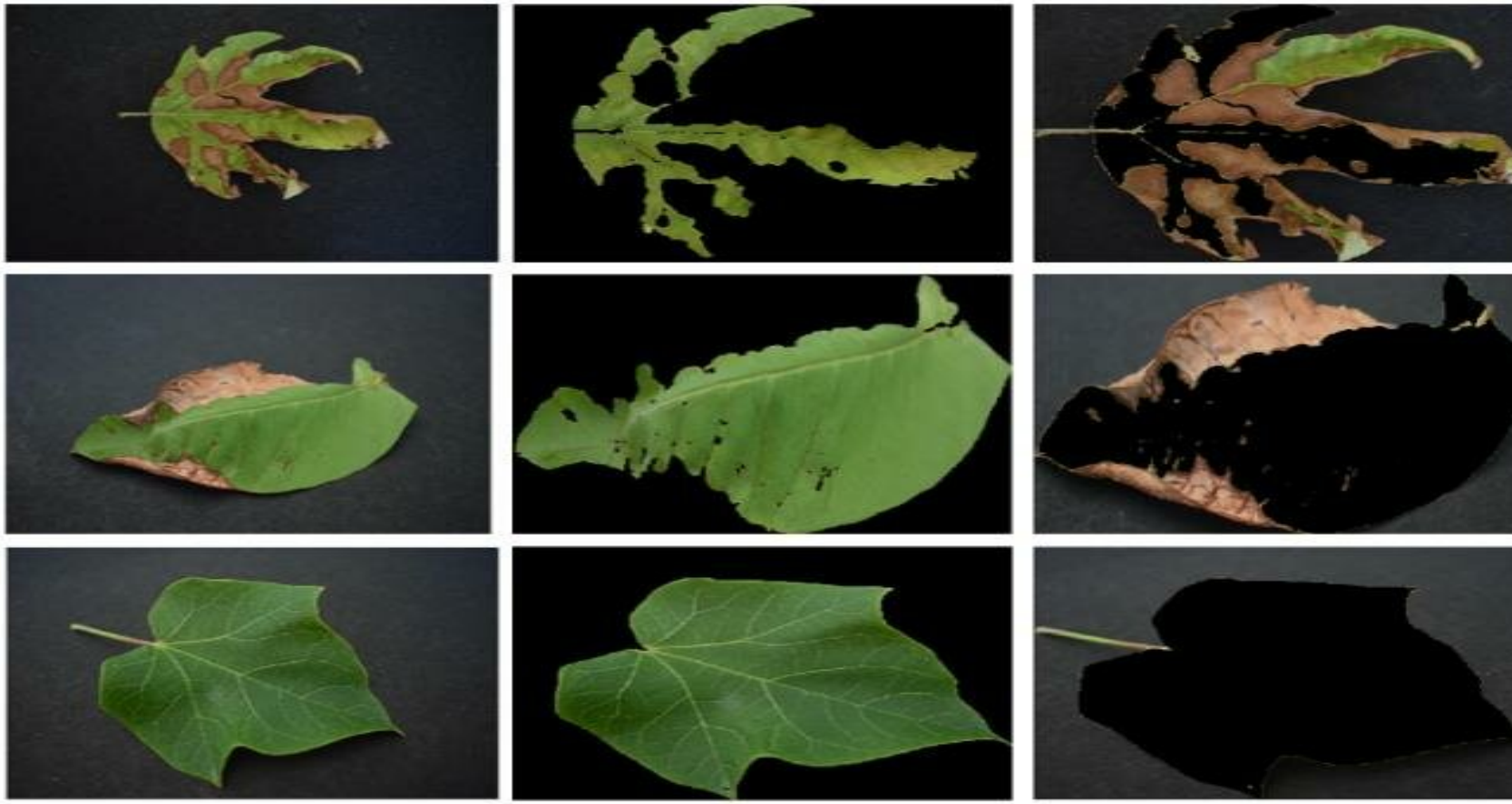
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```
Welcome to the Plant Disease Chatbot. Type 'exit' to quit.
User: What are common leaf diseases in corn?
Bot: Based on the following plant disease information: 5, 5, 5, 5, 5. Answer the question: What are common leaf diseases in cor

5 - Leaf blight
5 - Rust
5 - Powdery mildew
5 - Fungal infections
5 - Root rot

The common leaf diseases in corn are:

* Leaf blight
* Rust
* Powdery mildew

Note: The answer is based on the information provided in the question.
```



Future Work

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Components of the system:-

- Voice-based chatbot interaction.
- Real-time database updates from agricultural research.
- Mobile app development for farmers.
- Build a dataset with geo-location and weather data to improve predictions.



Conclusion

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- The multimodal chatbot for plant disease recognition combines FAISS-based text retrieval, CNN-based image classification, and LLaMA 2 for generating intelligent responses. LLaMA 2 + FAISS + SAM = Robust, scalable solution. This solution can help farmers and researchers improve crop health and yield. Future improvements may include support for more plant species, multilingual support, and real-time applications.



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A. Abbas, S. Jain, M. Gour, and S. Vankudothu, "Tomato plant disease detection using transfer learning with c-gan synthetic images," *Computers and Electronics in Agriculture*, June 2021.